Case Study 5

Analysis of real estate prices using Zillow, Realtor.com and FRED Data

Previous Assignment (Case Study 4)

- How did the Federal Reserve's monetary tightening from 2021–2026 propagate through consumer auto financing, bank lending standards, and durable-goods manufacturing output?
- Frequency and horizon: US monthly and quarterly series from 2018 – 2026 (pre-trend, shock window, transmission period)
- Effect
  - Auto credit and loan outcomes
  - Consumer credit and bank lending
  - Manufacturing (durable goods activity) and auto production

Background

- Zillow Research provides comprehensive real estate data, including market trends
  - Home Value Index
  - Home Value Forecast
- Realtor.com maintains data on housing inventory and market hotness

Assignment (Case Study 5)

- Synthesize the data from Zillow, Realtor.com and FRED to
  - Determine the housing market trend and create a prediction model for house prices in a neighborhood for the next 1 – 3 years
  - What should be the average household income to afford a 2500 sq ft house in the neighborhood that you are exploring?
- Choice of neighborhood
  - You can use the zip code(s) of the city where you come from
  - Alternately, you can explore some of the Bryan/College Station neighborhoods (zip codes: 77802, 77803, 77840, 77845)
- **Each team has to think about what additional items/features you will explore to strengthen your model**
  - **write this in your plan and incorporate in your case study**
- Form your team using the link on Canvas

Setting

- You are a risk analyst for Bank of Amarnath
- Bank of Amarnath has minimal presence in your zip code
- Should the bank offer more home loans in your zip code?

